Evaluation & Testing

Executive summary | Round 1 ran with all four teams: answers are right when the evidence is found, and each remaining gap has a named owner

105/124
questions where RAG's search finds all the evidence (85%)

89/94
numbers right in RAG's answers (0.947), scored by our graders

12/12
unanswerable questions correctly refused by RAG's reader

63 → 26
of 93 number questions: GraphRAG has the right number in its top 10 → ranks it first

What we did

- Ran round 1 with RAG (twice) and GraphRAG (both searches), and reproduced every score from their files
- Released Gold Set v2.2: the same 155 questions with fairer grading
- Closed all 11/5 feedback and connected the Agents' new API to our grader

What we found

- RAG finds and reads numbers well; 13 of its 16 declined questions are outside its reach today
- GraphRAG's gap is ranking and scope, not data: its graph search finds 10 of the 20 answers it holds
- Misspelled company names break the scope check (98% → 19%) unless matched loosely (94%)

What we need next week

- **RAG:** run our narrative judge
- **GraphRAG:** ranking fix; Exhibit 21 in the graph
- **Agents:** first run of the 155 questions
- **Data:** Microsoft's FY2024 and FY2025 10-Ks

So what: Round 1 gives every team a measured baseline on the same 155 questions, so next week we measure the fixes instead of debating them.

Source: round-1 files from RAG (11/9, 11/11) and GraphRAG (11/11), re-scored by Eval on Gold Set v2.2 (155 questions). Detail on the following slides; definitions in the appendix

Evaluation & Testing

Feedback from 11/5 | All seven action items are closed, each with a script and a result file in the Week 8 folder

Action item	What we did	Status
Run round 1 in week 8	RAG (11/9, 11/11) and GraphRAG (11/11) ran our scoring; we reproduced both	Done
Compare RAG and GraphRAG side by side; show where each does better	84 Microsoft 10-Q questions through each team's own code, plus 9 relationship questions	Done
Label the routing chart; define the keyword router and unneeded calls	Axes titled; both terms defined on the slide with a worked example	Done
Coverage vs accuracy; match each number to its method	Corrected: every tool 95%, keyword router 83%; coverage is not answer accuracy	Done
Concrete routing examples; where the router sits	Three real questions walked through; the keyword router is our baseline, not part of the pipeline	Done
State that results are on the current 155 questions only	Stated on every slide that reports a rate	Done
Test typos and messy wording	427 rewrites of the 155 questions, scored like the originals	Done

How to read every number

All figures use the current 155 questions (Gold Set v2.2).

104 finance data, 33 text, 9 relationship, 8 no tool, 1 both. Microsoft has 92 of them.

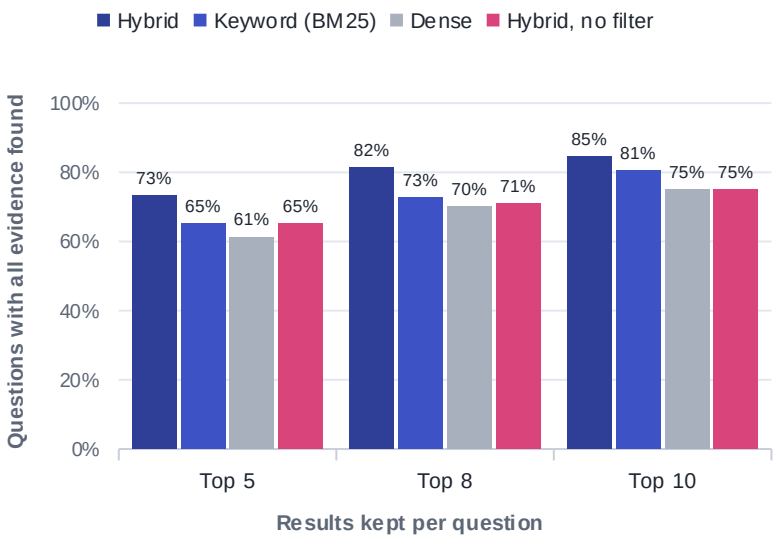
Small groups are examples, not estimates. Rates will move as questions are added.

So what: Two corrections matter most for reading last week's chart: the bars show coverage, not accuracy, and the 95% is calling every tool, not the keyword router.

Source: 11/5 meeting minutes (#6) and mentor feedback; Week 8/README.md maps each item to its script and result file

Evaluation & Testing

Round 1: RAG retrieval | RAG's hybrid search finds all the evidence for 105 of 124 questions, and every miss has a named fix



83/93

number questions found; text questions 22/31

34/37

on the nine other companies: adding them did not hurt retrieval

Why 19 were missed	n	Fix and owner
Cross-company ranking: one search cannot hold ten companies	3	Agents: one search per company
Our own anchors were wrong (four questions)	4	**Eval: fixed, with all 22 similar anchors**
Needs the 10-K and the 10-Q together	3	Agents: one search per form
Found by both methods; merging pushed it past 10	6	RAG: merge weights, keep more results
Missed by all three methods	3	RAG: ranking

So what: A wrong-period figure sits in the same top 8 for 90% of the 42 number questions that have one, so the reader must be told which column to take.

Source: RAG team, round-1 retrieval run 11/9 with Eval's scoring rules (Week 8/data/round1_rag_2026-11-09.json). Retrieval only; a hit must come from the right filing

Evaluation & Testing

Round 1: RAG answers | With the evidence found, RAG's reader gets 89 of 94 numbers right and refuses all 12 unanswerable questions

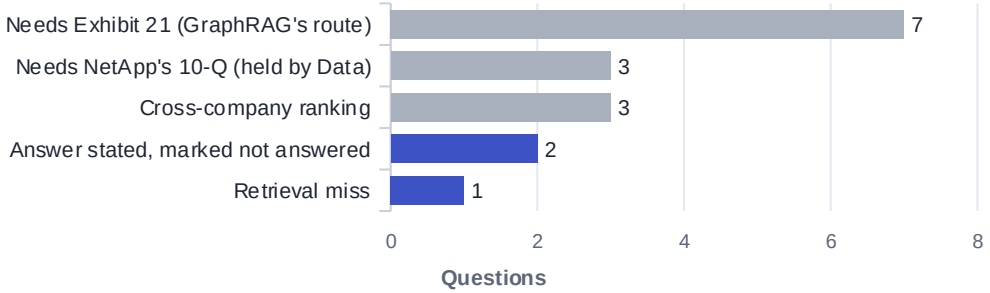
89/94
numbers right (0.947); 87/94 before our grading fixes

12/12
unanswerable questions refused; balanced refusal score 0.944

6/7
false-premise questions corrected instead of answered as asked

58/59
lookups show the number in a cited passage; all 185 citations exist

Why 16 answerable questions were declined (blue = RAG's own to fix)



How we checked

- Reproduced RAG's own figures exactly on its subset: numbers 0.935, refusal 0.977, unanswerable 10/10
- RAG's three grading notes were right; all are fixed in Gold Set v2.2 (appendix)
- 28 of 30 narrative questions were answered; scoring them waits for the LLM judge, which is built and needs an API key

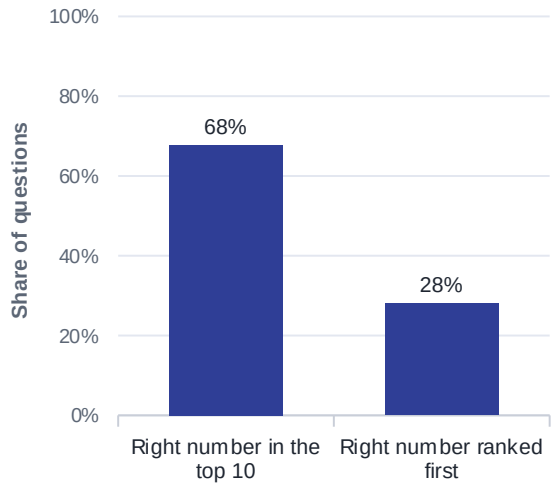
So what: Only 3 of the 16 declined questions are RAG's own to fix; the rest wait on Exhibit 21, one held filing and per-company search. Narrative scores come next.

Source: RAG's 155 answers (11/11) graded by Eval, Week 8/results/answers_rag_reader_round1.json. Gold Set v2.2; RAG's answers and passages stay private

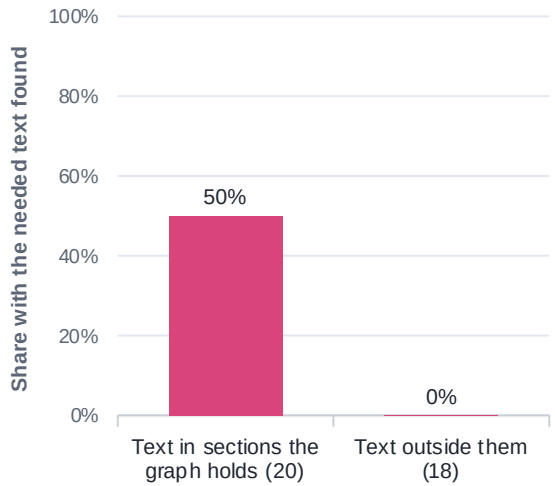
Evaluation & Testing

Round 1: GraphRAG | The right number is already in the top 10 for 63 of 93 questions; ranking it first (26 today) is the open step

Number search: 93 number questions



Graph search: 38 text questions



What we checked

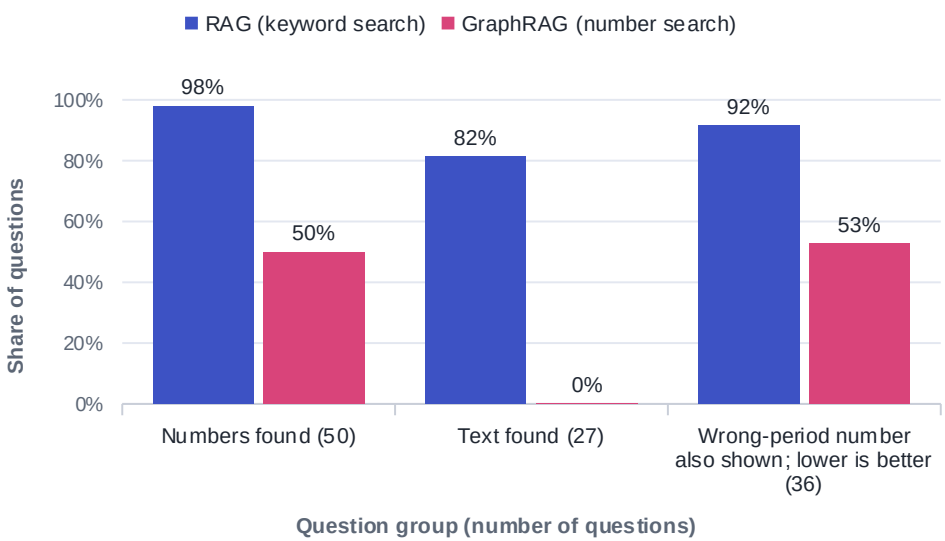
- Our rerun on GraphRAG's own 20-filing store matches its results on all 155 questions
- Its fix #70, merged the same day, removed all 10 errors on cross-company questions
- The graph holds five section types; 10 more answers sit in the notes to the statements, and 8 in Exhibit 21 or NetApp's 10-Q
- Dropped database connections are retried, never scored against GraphRAG

So what: Two levers: rank by period and break ties (GraphRAG's own issue #68), and add the notes to the statements to the graph. The data is already there.

Source: GraphRAG team, round 1 on its Aura graph and 20-filing store (11/11), re-run by Eval: Week 8/results/graphrag_round1_check.json. Retrieval only

Evaluation & Testing

RAG vs GraphRAG | On the same filing RAG finds more of the evidence, GraphRAG is more exact about the period, and relationship questions need GraphRAG



	Better at	Evidence
RAG	Finding the right number	49/50 vs 25/50
	Text and explanations	22/27 vs 0/27: number search holds numbers only
GraphRAG	Exact periods on numbers	Wrong-period number shown 53% vs 92%
	Relationship questions	9/9 exact by walking its subsidiary rows; RAG does not index Exhibit 21

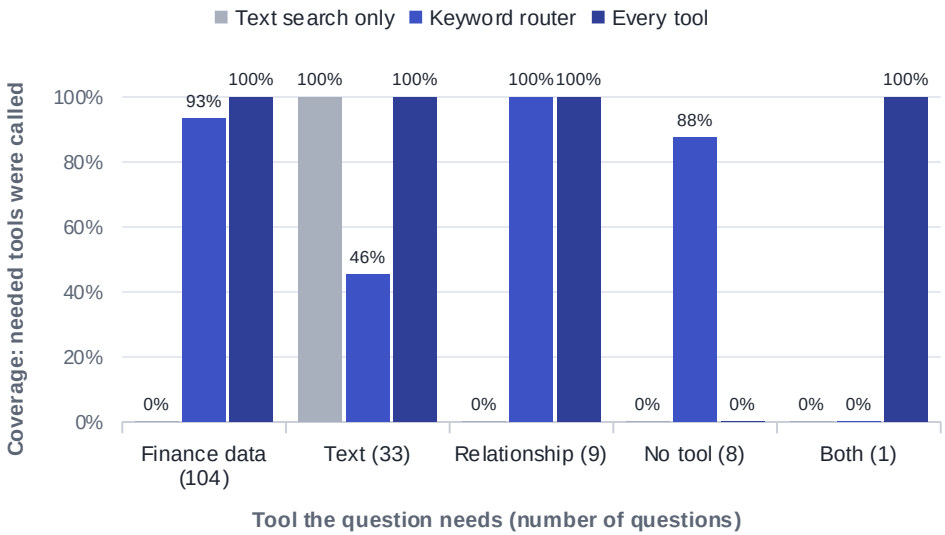
Neither search returns an empty result on the 7 unanswerable Microsoft questions, so the decision to abstain belongs to the Agents' evidence check.

So what: Route relationship questions to GraphRAG and explanations to RAG. Numbers can use either once GraphRAG's ranking fix lands, with its period labels as the check.

Source: Week 8/results/side_by_side.json: the same 84 Microsoft 10-Q questions through each team's own code, top 8 results. Retrieval only, no reader model

Evaluation & Testing

Routing | Coverage is not accuracy: calling every tool covers 95% of questions at 3 calls each, the keyword router 83% at about 1



Coverage = share of questions where every tool the question needs was called. **It is not accuracy:** calling the right tool does not make the final answer right.

Method	Coverage	Calls per question	Unneeded calls
Text search only	21%	1.00	0.78
Keyword router	83%	1.05	0.26
Every tool	95%	3.00	2.05

Unneeded call = a tool called that the question does not need. Total revenue needs finance data only, so calling every tool makes 3 calls and 2 are unneeded.

So what: Coverage rewards calling more tools, so always read it next to calls per question. Answer accuracy is measured separately, in round 1.

Source: Week 8/results/routing_baselines.json, 155 questions (Gold Set v2.2). Needed tools are Eval's labels, for the Agents to review. 'Both' and 'No tool' rest on 1 and 8 questions

Evaluation & Testing

Routing walkthrough | Three real questions show the trade-off: calling every tool always covers but wastes two calls, while keyword rules are cheap but can slip

1. A number from a statement

"What was Microsoft's total revenue for the three months ended March 31, 2026?"

Needs: finance data

Method	Calls	Covered	Unneeded
Text only	RAG	no	1
Keyword	Finance data	yes	0
Every tool	Finance data, RAG, GraphRAG	yes	2

Keyword router: number word 'total' → finance data

2. A relationship question

"How many subsidiaries does Micron list in Exhibit 21, and how many are outside the US?"

Needs: GraphRAG

Method	Calls	Covered	Unneeded
Text only	RAG	no	1
Keyword	GraphRAG	yes	0
Every tool	Finance data, RAG, GraphRAG	yes	2

Keyword router: relationship word 'Exhibit 21' → GraphRAG

3. Where keyword rules slip

"What does revenue allocated to remaining performance obligations include?"

Needs: RAG text search

Method	Calls	Covered	Unneeded
Text only	RAG	yes	0
Keyword	Finance data	no	1
Every tool	Finance data, RAG, GraphRAG	yes	2

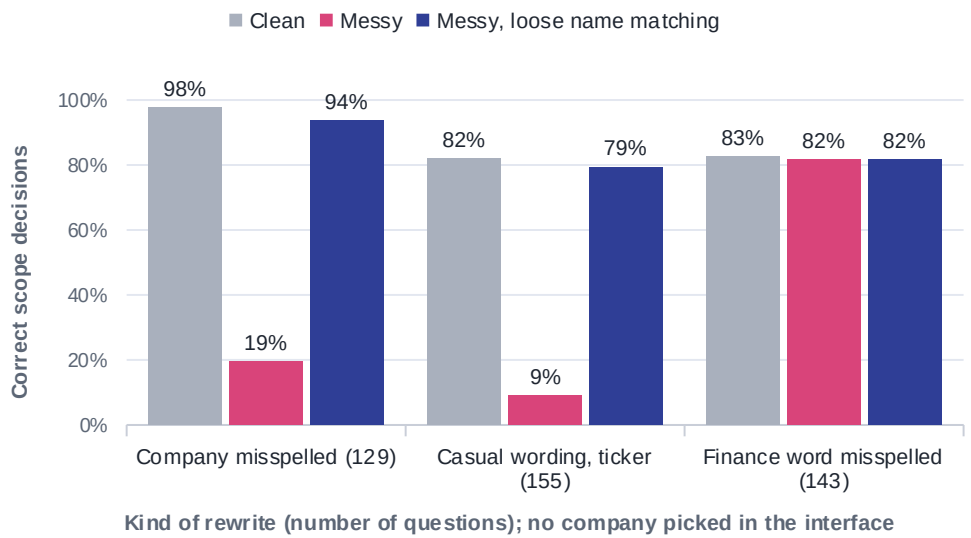
Keyword router: number word 'revenue' → finance data

So what: The keyword router is Eval's baseline, not part of the pipeline. The Agents' orchestrator does the real routing and can be scored on the same 155 questions with one command.

Source: Week 8/results/routing_examples.json; seven worked examples in ROUTING_EXAMPLES.md. For the Agents: scripts/run_routing_baselines.py --router name=module.py:route

Evaluation & Testing

Messy wording | Misspelled company names cut correct scope decisions from 98% to 19%; matching names loosely restores 94%



Effect on search (Microsoft 10-Q questions)

Rewrite	RAG search	GraphRAG search
Finance word misspelled	91% → 80%	30% → 17%
Company misspelled	93% → 93%	37% → 37%
Casual wording	92% → 91%	32% → 32%

Why

- **Names:** the check knows exact names only, so 'Microsfot' or 'msft' becomes 'which company?'
- **Finance words:** keyword matching does not forgive 'revnue'
- **Search ignores name typos** because the company arrives as a filter

So what: Two inexpensive fixes for the Agents: match tickers and near-miss names before routing, and correct spelling before search.

Source: Week 8/results/messy_wording.json: 427 rewrites of the 155 questions with the same meaning. Scope check and router are Eval's reference rules, not the Agents' orchestrator

Evaluation & Testing

Agents API v0.2 | The new /query fields plug straight into our grader; round 1 says fallback has 1 question to rescue and 139 to protect

What each API field lets us score

API field	What we score
status	The decision: answer when answerable, abstain when not
evidence	The needed figure or passage is in the evidence
citations	Each link points to the right filing
filters	Company and period were detected correctly
evidence_status, fallback flags	When fallback to GraphRAG fires, and whether it helps
answer (empty in v0.2)	Same grader as RAG's reader, once it lands

What round 1 expects from fallback, per question

Case	Questions
RAG already answers: fallback must not break it	**127**
Unanswerable: must stay “no evidence”	**12**
Needs Exhibit 21, not in the graph yet	7
Cross-company: one search per company	3
Needs NetApp's 10-Q, held by Data	3
Answer was already in RAG's evidence	2
GraphRAG can rescue today	**1**

So what: Our runner and grader are ready and tested against the contract's response format. The next step is the Agents' first run on the 155 questions.

Source: Agents' API Contract v0.2 (11/11); Week 8/AGENTS_API_W8.md; Week 8/data/fallback_cases_w8.csv, built from RAG's and GraphRAG's round-1 results

Evaluation & Testing

Next steps | One ask per team for next week, each tied to a round-1 result

Team	Ask for next week	Round-1 result behind it
RAG	Run our narrative judge with your key; mark answers that state the figure as answered	28 narrative answers not yet scored; 2 answers stated but flagged as declined
GraphRAG	Rank by period and break ties (#68); add the notes to the statements; load Exhibit 21	Right number in the top 10 for 63/93, first for 26; 10 text answers sit in the notes; 7 questions wait on Exhibit 21
Agents	Run the 155 questions through /query v0.2; search once per company; match tickers and near-miss names	Runner and grader are ready; 3 rankings need per-company search; names 98% → 19%
Data	Load Microsoft's FY2024 and FY2025 10-Ks; release NetApp's 10-Q when its review is done	Multi-year questions need them (joint ask with GraphRAG); 3 questions wait on NetApp
Eval	Hand-label narrative answers to validate the judge; grade the Agents' first run; add text anchors to the last 40 lookups	The judge counts only after it agrees with our hand labels

Thank you: RAG ran round 1 twice and GraphRAG ran both searches, and both traced every miss. That is how we found and fixed our own errors.

Source: Week 8/FOR_TEAMS_W8.md (one section per team, with file paths) and COMMS_W8.md (replies sent to each team)

Evaluation & Testing

Appendix: Gold Set v2.2 | Same 155 questions and expected answers; six grading changes, three of them raised by RAG

Change	Questions	Why	Raised by
Precision asked = precision the filing prints	EV1-015, EV1-027	“$0.91” was failed for having two significant figures, which is all the filing prints	**RAG**
Every figure required on multi-part questions	7 questions, e.g. EV1-043	The question asks for three changes; the answer format held one value	**RAG**
Route moved from relationship to text	EV1R-002, EV1R-006	RAG answered both from the 10-K text; GraphRAG's review agreed	**RAG**
A second accepted wording	EV1R-005	The 10-K says Microsoft holds about 25% of OpenAI; the earlier 10-Q says 27%	Eval
One evidence item per required point	22 questions	Two sentences in one item could never match once RAG's index split them	Eval
Last text anchors added; two labels corrected	EV2-035/036/037/044; EV3-008/009	Anchors read from RAG's passages; two fiscal-calendar questions were labelled as relationship	Eval

So what: Numbers right moves from 87/94 to 89/94; the questions do not change. Earlier versions are kept, so earlier runs stay comparable.

Source: Week 8/GOLD_SET_V2.md and results/gold_v2_2_changes.json (38 questions touched; file hash 416304a91b4aa400)

Evaluation & Testing

Appendix: GraphRAG detail | Most number misses have a specific fix, and GraphRAG's Exhibit 21 rows match our own reads on 62 of 62

Where the 25 number misses on the Microsoft 10-Q come from

Cause	n	Fix
Right number found but ranked below 8; in 6, “three months” matches Microsoft 365 rows	13	Treat period words separately; break ties
Calculations that need several inputs	10	One call per input
Wording does not reach the concept	2	Two aliases (unearned revenue, share repurchases)

Hop counts (GraphRAG, PR #71): filled for 44 questions and left blank on purpose where the answer comes from finance data, which the graph does not hold. EV3-002 counts 2 hops in GraphRAG's design and 4 in ours: a design difference (country is a property, not a node), not a counting error.

9/9
relationship questions answered exactly by walking GraphRAG's subsidiary rows

62/62
Exhibit 21 rows agree with our own reads of the filings

14/50
right number at rank 1 on the Microsoft 10-Q

6/36
wrong-period number at rank 1; GraphRAG flagged this itself

So what: On GraphRAG's own 20-filing store the same comparison gives 24/50 against 25/50 on ours, so the store does not explain the gap; ranking does.

Source: Week 8/results/side_by_side.json (number_miss_diagnosis, part_b), graphrag_round1_check.json, hops_v2_summary.json; GraphRAG PR #71 and Exhibit 21 branch c21fe89

Evaluation & Testing

Appendix: Definitions | How to read the measures in this deck

Term	Meaning	In this deck
Gold Set v2.2	Our 155 questions with expected answers, on 10 companies and 21 filings	143 answerable, 12 unanswerable
Evidence found	Every figure or passage the answer needs is in the first k results, from the right filing	105/124 at 10 results
Numbers right	The stated figure matches the filing at the precision it is written	89/94
Balanced refusal	Average of refusing unanswerable questions and not refusing answerable ones	0.944
Coverage	Every tool the question needs was called; it says nothing about the answer	95% for every tool
Unneeded call	A tool called that the question does not need	2.05 per question for every tool
False premise	The question assumes something untrue; a good answer corrects it	6/7 corrected
Fallback	The Agents try GraphRAG when RAG's evidence is weak or empty	1 question to rescue today

Reproducible: Every number re-runs with one command (scripts/run_all.sh, 100 tests). Other teams' code is pinned: RAG 5bfafeb, GraphRAG 21f0795, Data 7040b8a.

Source: Week 8/README.md; kpmg_eval/metrics. No LLM is used in Eval's own results; the teams' round-1 runs are re-scored from the files they sent